

Four clinical ECG sources

PTB-XL | Georgia | Chapman | Ningbo

77,333 records -> 66,379 retained, 13 harmonised labels

Native 500 Hz ECG

quality indicators measured
before decimation

Anti-aliased resampling to 100 Hz

model input: 12 x 1000

RQ1. External generalisation

leave-one-source-out evaluation of six compact encoders

held-out predictions

RQ2. Reliability under source and quality shift

quality characterisation, prevalence-matched dAUPRC,
and selective reliability

externally evaluated models

RQ3. Deployment-preserved performance

int8 on STM32F411: Flash, SRAM, latency,
accuracy-resource trade-off

*quality
indicators*